Kiran Kishore Venkatesan

AI/ML Engineer Intern | B.Tech – CSE (Artificial Intelligence) | **Amrita Vishwa Vidyapeetham, Coimbatore**
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Summary

Final-year AI undergraduate and **AI/ML engineer** who turns research ideas into working, deployable systems. Hands-on across the full ML lifecycle (data preprocessing, model design, evaluation, and cloud deployment) in **Generative AI, Agentic AI, Large Language Models (LLMs), Computer Vision, and NLP**, with a focus on real-world impact in healthcare, accessibility, and developer tooling. Seeking an AI/ML Engineer Intern role to build reliable, production-grade AI products.

Education

Amrita Vishwa Vidyapeetham, Amrita School of AI *2023 – 2027*
B.Tech – Computer Science and Engineering (Artificial Intelligence) CGPA: **8.06/10**

Notable Achievements

Guinness World Record Holder | *Largest Electronic Keyboard Ensemble (440 participants)* *2017*
4-time Hackathon Finalist | *Civic Stack (IIT Delhi), AWS AI for Bharat, Technova, 2nd SmallAI*

Experience

Latlon Technologies Pvt. Ltd. *May 2026 – Jun 2026*
AI/ML Intern Coimbatore, India

- Co-developed an **autonomous, LLM-powered developer environment** (Agentic AI) that analyzes and extends codebases.
- Automated **legacy-codebase modernization**, pairing AI code generation with build and validation steps to cut manual effort.

NetKathir Technologies Pvt. Ltd. *Apr 2025 – May 2025*
AI Developer Intern Puducherry, India

- Developed an **AI-powered 3D medical conversational avatar** for real-time, voice-based patient interaction.
- Integrated a **3-stage speech pipeline** (speech-to-text, NLP response generation, text-to-speech) to answer patient queries.

Projects

Lazarus – AI-Powered Legacy Codebase Modernization Platform *Jan 2026 – Mar 2026*
Agentic AI | AWS Bedrock, Next.js, FastAPI, Docker, AWS Lambda, DynamoDB, S3

- Built a **5-agent AI pipeline** (Inspector, Architect, Builder, Deployer, Validator) on **AWS Bedrock** for **AST-based analysis, dependency-graph extraction, and automated code migration** of legacy repositories.
- Engineered **serverless orchestration** on Lambda, Step Functions, and API Gateway with a **self-healing loop** that parses CloudWatch logs and patches failures, reaching a **97% build success rate in roughly 8 minutes**.

Skeleton-Based Sign Language Recognition using GCN–TCN *Oct 2025 – Apr 2026*
Deep Learning, Computer Vision | PyTorch, MediaPipe, OpenCV, NumPy, Scikit-learn

- Achieved **85.71% test accuracy and 0.893 macro F1** on 10-class Indian Sign Language (ISL) sentence recognition with a 2-layer spatial GCN + 4-block dilated residual TCN, up from 50% for the full-landmark baseline.
- Built the **skeleton graph and spatial GCN** on 35 eigenvalue-selected landmarks (**53% fewer graph nodes, 78% of motion variance kept**), **resolving baseline overfitting**: train/test accuracy moved from 94.6%/50.0% to **90.2%/85.7%**.

UrbanPulse – AI-Based Civic Issue Monitoring & Decision Support System *Dec 2025 – Mar 2026*
AI/ML Engineer | Roboflow CNN, DistilBERT, Grad-CAM, Kalman Filter, AHP, React Native, Supabase

- Built the **AI verification layer**: a Roboflow CNN with **Grad-CAM** validates issue photos, **DistilBERT** scores urgency from text and Sarvam voice input, and a Puppeteer scraper ingests issues from 3 social platforms, with spoofing checks.
- Designed **AHP-weighted priority scoring** over 4 signals (image, urgency, proximity, social) and a **Kalman-filter severity forecast** for predictive maintenance, surfaced as 3-km heatmaps and an admin XAI dashboard.

Technical Skills

- **Languages:** Python, C++, C, SQL, Java, JavaScript, HTML/CSS
- **AI/ML:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs), Agentic AI, Graph Neural Networks (GCN), Prompt Engineering, Model Evaluation
- **Libraries & Frameworks:** PyTorch, NumPy, Scikit-learn, Hugging Face Transformers, TensorFlow Lite, OpenCV, MediaPipe, YOLOv11
- **Cloud & DevOps:** AWS (Bedrock, Lambda, Step Functions, API Gateway, DynamoDB, S3), GCP, Docker, Git/GitHub
- **Backend & Web:** FastAPI, REST APIs, Next.js, React Native, Supabase (PostgreSQL)

Certifications

AWS Certified AI Practitioner (AIF-C01) | *Amazon Web Services*
AWS Certified Cloud Practitioner (CLF-C02) | *Amazon Web Services*
Blockchain Basics | *Cyfrin Updraft*